

FRONT END DEVELOPMENT FRAMEWORKS AND UI ENGINEERING (25CS1201E)
2025-26, Term-III
Project Problem Statement Submission Form

Sec No-	Team No-	Project Title- Airlink – Smart Airport Operations & Flight Management System
Team Lead Roll Number	2520090156	
Team Lead Name	THIRUNAGARI SAIKRISHNA	
Team Member Roll Number		
Team Member Name		

1. Abstract

(Summarize the project idea, problem background, and proposed solution. Not more than 250 words.)

The Flight Status Tracker System is a web-based application that helps users check real-time flight information easily. Many passengers face problems due to delays, gate changes, or lack of updated flight details. This system is designed to provide accurate and up-to-date flight status in a simple and user-friendly way.

Users can search for a flight using its flight number and view details such as departure time, arrival time, gate number, and current status (on-time, delayed, or cancelled). The system also allows users to subscribe to specific flights and receive notifications about any updates or changes.

The application includes a dashboard that displays important flight information and supports filtering by airline. It also provides live updates using API integration. An admin panel is included to manage flight data and users.

The system is developed using modern web technologies like HTML, CSS, JavaScript, React.js, Node.js, and REST APIs. It ensures fast performance and responsive design for both desktop and mobile users.

Overall, this project makes it easier for passengers to stay informed and reduces confusion during travel.

2. Module Mapping

(Provide details of how modules are divided among team members.)

Module Name	Assigned Roll Number(s)	Signature
Authentication & Authorization	2520090156	
Dashboard / Home Screen	2520090156	
Core Feature A (Flight Search & Status Tracking)	2520090156	
Core Feature B (Live Tracking & Subscription)	2520090156	
Form Handling	2520090156	
Data Display	2520090156	
Notifications & Alerts	2520090156	
File Handling	2520090156	
API Integration	2520090156	
Admin Panel	2520090156	

3. Module Details

(Explain the responsibility of each student concerning their assigned module.)

Module 1 — Authentication & Authorization

Handles user registration, login, and secure access using JWT/session management. Supports role-based access for users and admins.

Module 2 — Dashboard / Home Screen

Displays an overview of flight statuses with KPI cards such as total flights, delayed flights, and active subscriptions. Provides quick navigation to search and tracking features.

Module 3 — Core Feature A (Flight Search & Status Tracking)

Allows users to search flights using flight numbers and view details like departure, arrival time, gate number, and current status (on-time, delayed, cancelled).

Module 4 — Core Feature B (Live Tracking & Subscription)

Provides real-time updates using polling or WebSockets. Users can subscribe to specific flights and receive updates on status changes.

Module 5 — Form Handling

Implements validated forms for login, registration, and flight search with proper error messages and success feedback.

Module 6 — Data Display

Displays flight data using tables or cards with sorting, filtering (by airline/status), and pagination for better user experience.

Module 7 — Notifications & Alerts

Shows real-time alerts for delays, gate changes, and cancellations using toast notifications, modals, and an in-app notification center.

Module 8 — File Handling

Allows users/admins to download flight reports or booking details as PDF files. May include upload of documents (optional for admin use).

Module 9 — API Integration

Integrates external flight APIs (like AviationStack/OpenSky) using Axios. Handles loading states, success responses, and error handling. APIs are tested using Postman.

Module 10 — Admin Panel

Provides administrative features such as managing flight data, monitoring users, viewing reports, and controlling subscriptions.

4. Frameworks / Platforms / Software Used

(List the frameworks, libraries, platforms, or tools you plan to use.)



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The Flight Status Tracker System is developed using a full-stack approach with modern technologies.

- **HTML5, CSS3, JavaScript (ES6+)**
- **React.js, React Router**
- **Node.js, Express.js**
- **Axios (API integration)**
- **Bootstrap / Tailwind CSS**
- **MySQL (Database)**
- **Postman (API Testing)**
- **External Flight APIs (AviationStack / OpenSky)**

Guide Name and Signature (remarks if any) _____